15-10-2026

Subject : Lab 4

Department : BSCS (1st semester)

Section : C

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Notes

Date:

LAB 04 — C++ Programming

Task 1: Print Positive or Negative Number

Question:

Write a program that takes a number from the user and prints whether it is positive or negative.

Solution:

```
#include <iostream>
using namespace std;
int main(,
{
  int num;
  cout << "Enter a number: ";
  cin >> num;
  {
  if (num > 0)
  {
  cout << "The number is positive." << endl;
  }
  else if (num < 0)
  {
  cout << "The number is negative." << endl;
  }
  else
  {
  cout << "The number is zero." << endl;
  }
  return 0;
  }
```

Rana Notes

Task 2: Check Positive, Negative, or Zero

Question:

Write a program to check whether an integer is positive, negative, or zero.

Solution:

```
#include <iostream>
using namespace std;

int main()
{
    int num;
    cout << "Enter an integer: ";
    cin >> num;

    if (num > 0)
    {
        cout << "Positive number" << endl;
    }
    else if (num < 0)
    {
        cout << "Negative number" << endl;
    }
    else
    {
        cout << "Zero" << endl;
    }

    return 0;
}
```

Example Output:

```
Enter an integer: -5
Negative number
```

Rupalee Notes

Task 3: Calculate a Student's Grade

Question:

Write a program to calculate a student's grade according to the following criteria.

Marks	Grade
75 or above	A
60 to below 75	B
45 to below 60	C
Below 30	D

Solution:

```
#include <iostream>
using namespace std;

int main()
{
    int mark;

    cout << "Enter your marks: ";
    cin >> mark;
```

Royale Notes

```
    if (mark < 0 || mark > 100)
    {
        cout << "Invalid marks." << endl;
    }
    else if (mark >= 75)
    {
        cout << "Grade A" << endl;
    }
    else if (mark >= 60)
    {
        cout << "Grade B" << endl;
    }
    else if (mark >= 45)
    {
        cout << "Grade C" << endl;
    }
    else if (mark < 30)
    {
        cout << "Grade D" << endl;
    }
    else
    {
        cout << "Grade not specified in the question." << endl;
    }
    return 0;
}
```

Example Output:

```
Enter your marks: 82
Grade A
```

Note: The question does not specify a grade for marks from 30 to below 45.

Rayate Notes

Task 4: Find the Largest Number Using Nested If-Else

Part A: Find the Largest of Three Numbers

Question:

Write a program to find the largest of three numbers using nested if-else statements.

Solution:

```
#include <iostream>
using namespace std;

int main()
{
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    if (a >= b)
    {
        if (a >= c)
        {
            cout << "Largest number = " << a << endl;
        }
        else
        {
            cout << "Largest number = " << c << endl;
        }
    }
    else
    {
        if (b >= c)
        {
            cout << "Largest number = " << b << endl;
        }
        else
        {
            cout << "Largest number = " << c << endl;
        }
    }
    return 0;
}
```

Example Output:

```
Enter three numbers: 12 25 18
Largest number = 25
```

Rayate Notes

Part B: Check Whether a Number Is Positive and Count Its Digits

Solution:

```
#include <iostream>
using namespace std;
int main()
{
    int num;
    int digits = 0;
    cout << "Enter a number: ";
    cin >> num;
    if (num > 0)
    {
        while (num > 0)
        {
            digits++;
            num = num / 10;
        }
        cout << "The number is positive." << endl;
        cout << "Number of digits = " << digits << endl;
    }
    else
    {
        cout << "The number is not positive." << endl;
    }
    return 0;
}
```

Example Output:

```
Enter a number: 4567
The number is positive.
Number of digits = 4
```

Regular Notes